



UI / UX System for the Metalix frontend

How every screen, component and interaction should look and behave: a practical, illustrated guide for building the shadcn/ui frontend so that owners, accountants and yard staff understand it at first sight.

Dashboard

Good afternoon, Rajesh. Here's September so far.

Today

This week

This month

This FY

Stock in yard

286.5 t

▲ 31.520 t since 01/09/2026

Buy needed

38.8 t

3 materials are short

Open sales orders

14

212.300 t left to deliver

Delivered this month

₹1.96 Cr

▲ 8% vs Aug

What is inside

Each topic has a rules page, and every important screen has a mockup drawn with the real proposed colours, sizes and sample data. Red numbered markers on mockups point to the notes underneath.

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Sample data used throughout

Business "Metalix", today = 27/09/2026. Suppliers: Shree Ganesh Metals, Sharma Traders, Patel Scrap Co., Jai Ambe Steel, Noor Metal Mart. Buyers: Bharat Castings Pvt Ltd, Rajkot Alloys, Kutch Foundry Works. Materials: HMS 1, MS turnings, CRC bundle, Aluminium extrusion, Brass honey, Copper wire (Berry). All numbers are consistent with each other: stock sums to 286.450 t, every amount = tons × rate.

The system on one page

Every screen answers one of the owner's two daily questions: **"What do I need to buy?"** and **"What can I deliver?"**. The rest of this guide is these eight principles applied to each part of the app.

1

Numbers first, calm around them

Neutral greys plus one steel-blue accent. Colour appears only when something needs attention (amber = close to a limit, red = over a limit). Big, aligned, exact numbers.

2

One way to do each thing

One sidebar, one page layout, one table (that becomes cards on phones), one form pattern, one number formatter. Learn it once, use it everywhere.

3

Prevent errors, don't report them

The delivery form shows PO remaining, yard stock and supplier stock *before* saving, names the tightest limit and offers "Use max". Mistakes are hard to make.

4

Three clicks per core task

Smart defaults (today, last supplier, last rate), "Deliver" buttons right on the order row, inline "+ Add" in pickers, "Save & add another".

5

Plain trade words

"Left to deliver", "Stock with supplier", "Rate per ton". Sales POs are "Sales Orders"; statuses are Pending, Partly delivered, Completed, Cancelled.

6

Never a blank screen

Old rows stay visible while new data loads; skeletons only on the first load and only after 0.5 s; forms keep what users typed through every error.

7

Motion only as feedback

100–300 ms, never more than 400 ms. Animations confirm a press, a save or a change. No count-ups, shakes, confetti or parallax.

8

Built for the yard too

44–48 px touch targets, bottom navigation, drawers instead of dialogs, cards instead of tables, and a high-contrast "Sunlight" theme for outdoors.

The signature interaction

Delivery limits



Shree Ganesh Metals stock

9.200 t

Tightest limit



You can deliver up to

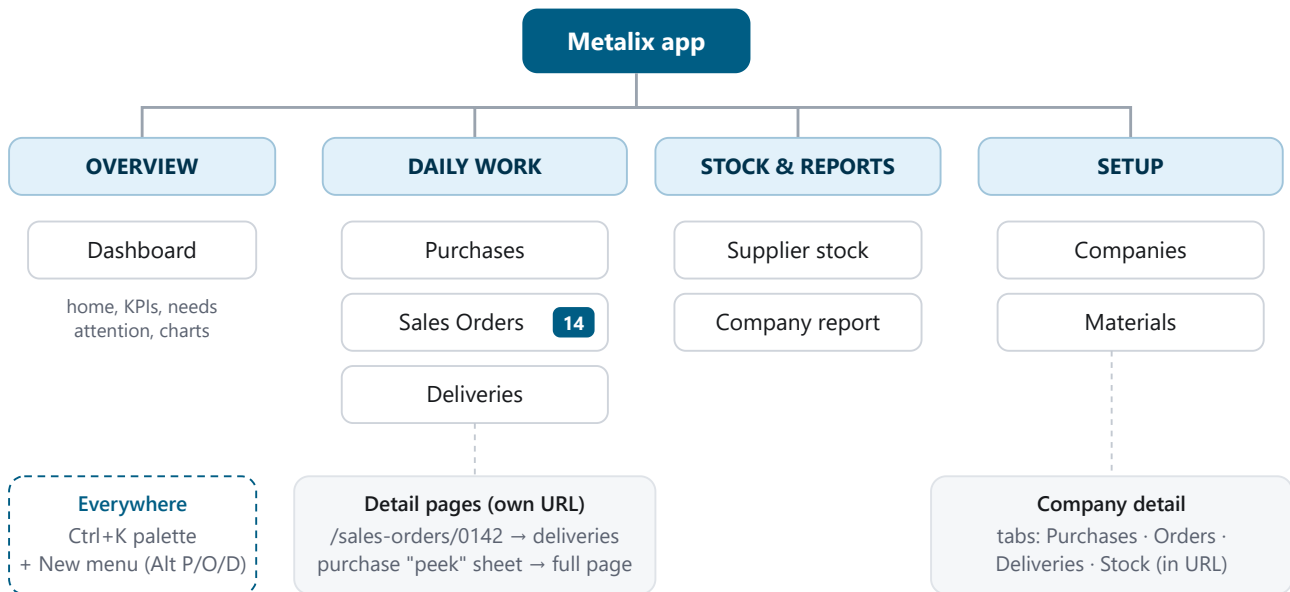
9.200 t

Limited by Shree Ganesh Metals stock

The backend already refuses any delivery bigger than the PO remaining, the yard stock or the chosen supplier's stock. The UI turns those three hidden rules into a visible panel that updates as the user types. Most "failed save" moments disappear. See pages 12–13.

Information architecture and navigation

Eight modules in four labelled groups, ordered by how often people use them. The sidebar is always visible on desktop: hidden menus made desktop users at least 39% slower in NN/g testing.



Say it the way the trade says it

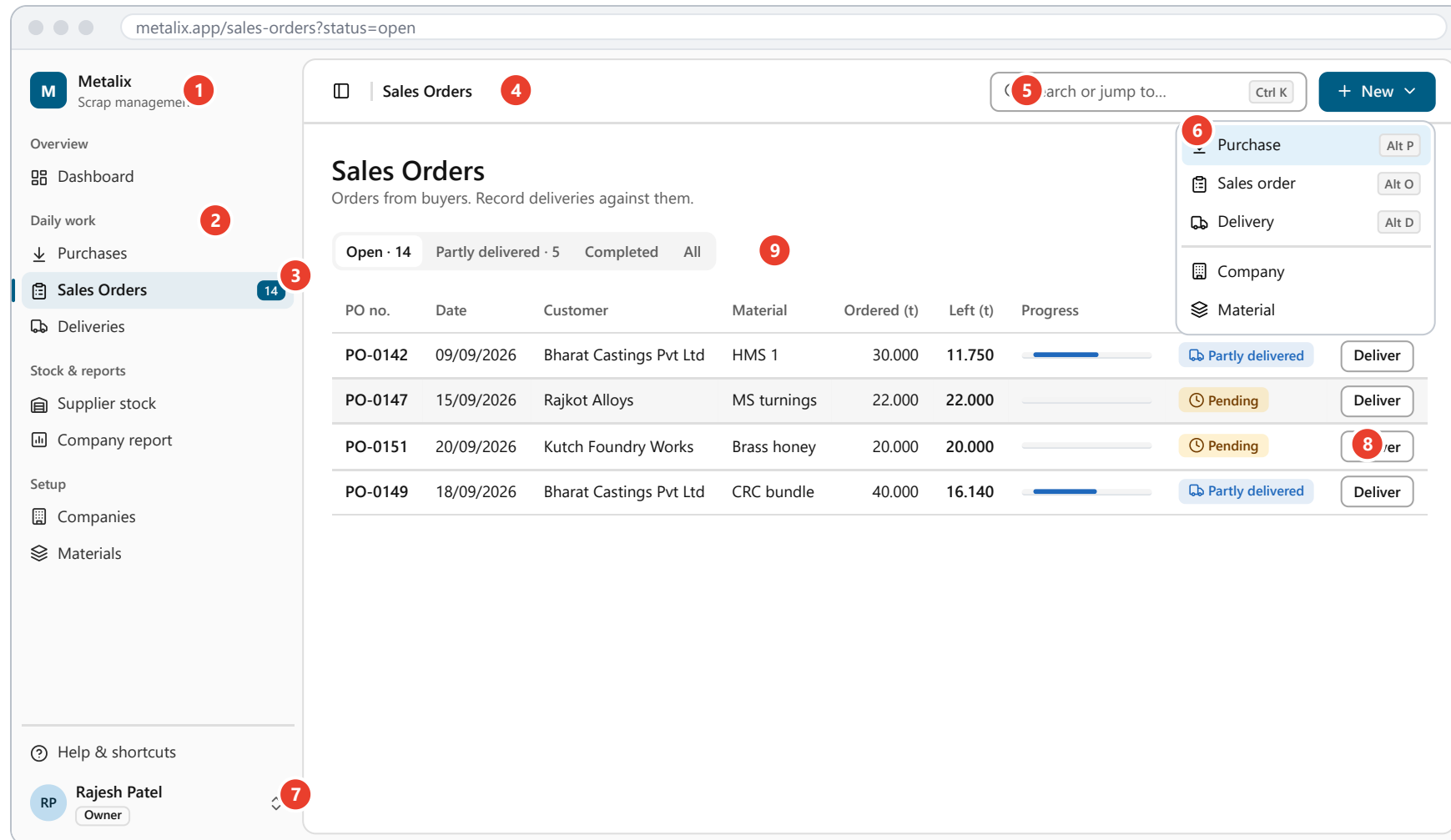
SYSTEM WORD	USE INSTEAD
Sales PO	Sales Order (numbers stay PO-0142)
PARTIALLY_SUPPLIED	Partly delivered
Balance qty	Left to deliver
Source company stock	Stock with supplier
Unit price	Rate per ton (₹/t)
Source company	Stock from (on the delivery form)

Rules

- Every item has an icon **and** a text label, never an icon alone.
- Sales Orders shows a count badge of open orders (14).
- **VIEWER role:** write buttons are hidden, not greyed out. A small "View only" badge in the user menu explains why. The server still enforces 403.
- Every filter, tab, sort and page lives in the URL, so Back works and a filtered view can be shared on WhatsApp.
- Help, shortcuts and a WhatsApp support link always sit in the same place (sidebar footer).

App shell: the frame around every page

shadcn Sidebar (collapsible to a 48 px icon rail, Ctrl+B, state remembered 7 days). Below 768 px it becomes a bottom bar (page 22).



- 1 Logo + business name. In the 48 px rail only the "M" square shows.
- 2 Four groups by frequency of use; icon + text label on every item.
- 3 Active item: tinted background, 3 px blue bar, `aria-current="page"`. Badge = open orders.
- 4 Sidebar toggle (Ctrl+B) and breadcrumb; detail pages read "Sales Orders > PO-0142".
- 5 Visible search button with a "Ctrl K" chip so occasional users discover the palette.
- 6 "+ New" creates anything from anywhere, with Tally-style Alt shortcuts. Hidden for VIEWER.
- 7 User row shows the role ("Owner" / "View only"); menu holds Theme incl. Sunlight, Reduce animations, Log out.
- 8 One visible "Deliver" button on open orders: the most frequent action, one click away.
- 9 Quick-view tabs instead of a status filter; exactly one primary action per page.

Page anatomy, containers and shortcuts

Every page is built the same way, and every create/edit form opens in the same kind of container. Users stop hunting for buttons after the first day.

One page anatomy

Header: ≡ · breadcrumb · search · + New

Primary

① Title · description · 1 primary action

search · date · supplier · material

② Filter bar (+ chips when active)

46 purchases · 412,380 t · ₹1,84,62,310.00

③ Summary strip (whole filtered set)

④ Content: DataTable / cards / stat cards

Showing 1–25 of 46 · 1 2 >

⑤ Pagination / footer actions

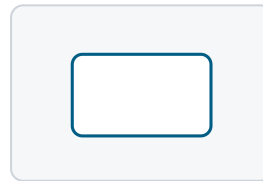
Padding 24 px (16 px on phones). Lists use the full width; forms stay ≈ 560–640 px wide; detail pages split 2/3 + 1/3 from 1024 px.

Which container for which job



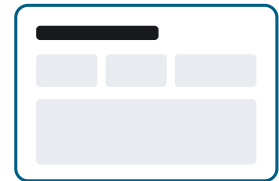
Right Sheet

create / edit records
512–576 px (768 delivery)



Small Dialog

add company from a picker;
AlertDialog = confirm only



Full page (own URL)

details with related records:
sales order, company, material

- Sheets keep the list visible behind them, so users never lose their place (NN/g: modals make people forget details they need).
- Sheets have URLs (/sales/new?poId=0142): refreshing never loses the form, and drafts are saved locally.
- On phones every Sheet and Dialog becomes a bottom **Drawer**.

Keyboard: Tally-speed for accountants

Ctrl K

Command palette: go to, create, search, recent

Alt P / O / D

New purchase / sales order / delivery (F9, F8 as aliases)

Alt C

Inside a picker: add the typed company or material

Enter · Ctrl ↵

Next field · save the form

/ · Esc · ?

Focus table search · close overlay · show all shortcuts

Command palette (Ctrl+K)

Q shree Esc

Companies

Shree Ganesh Metals Supplier · Rajkot

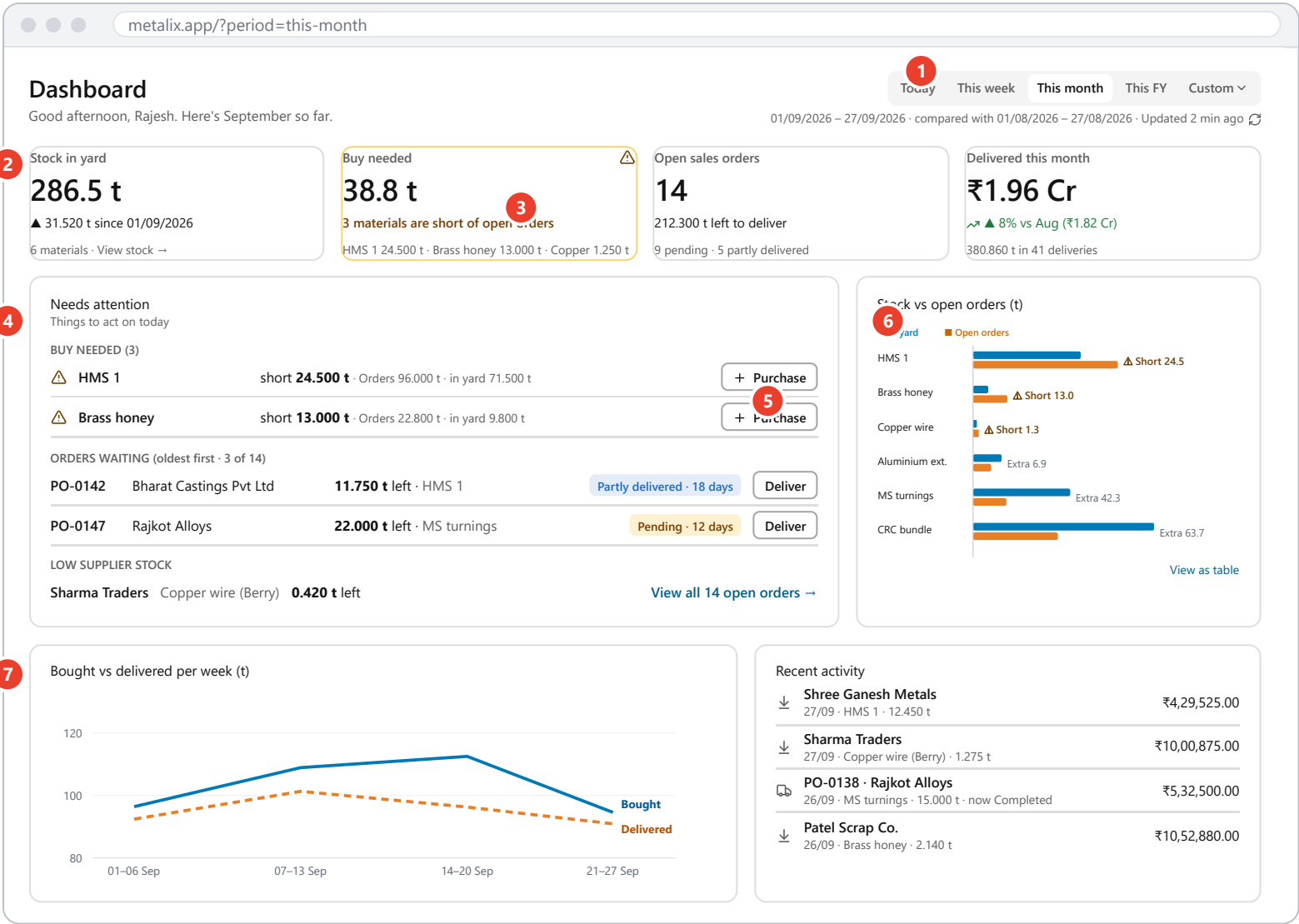
Create

↓ New purchase from Shree Ganesh Metals Alt P

Recent

PO-0142 · Bharat Castings Pvt Ltd opened 10 min ago

Dashboard: "what do I buy, what can I deliver" on one screen



- 1** **Period control** with Indian presets, resolved dates in words, comparison period and "Updated 2 min ago". Refetch every 60 s.
- 2** **Four KPI cards only.** Each has a comparison ("▲ 8% vs Aug") and links to the filtered list behind it.
- 3** **"Buy needed"** = open-order demand minus stock, per material. The most actionable number; the only card with colour.
- 4** **Needs attention:** materials to buy, oldest orders waiting, low supplier stock, each with its one-click action.
- 5** Inline actions open the Sheet with material / PO already filled in. Hidden for VIEWER.
- 6** **Horizontal bars** sorted by shortage, end-labelled "Short 24.5". Axis starts at zero. "View as table" for exact values.
- 7** Trend as two lines, **labelled at the line ends** (no colour legend), solid vs dashed as a second cue.

No pies, donuts, gauges or 3-D. KPI numbers never "count up". A margin card waits for a per-material rate endpoint.

Dashboard rules and data visualisation

A dashboard is the most important information "arranged on a single screen so it can be monitored at a glance" (Stephen Few). Every number needs context to show whether it is good or bad, and every number should lead somewhere.

The four KPI cards

CARD	VALUE	CONTEXT LINE	CLICK GOES TO
Stock in yard	286.5 t (tooltip 286.450 t)	▲ 31.520 t since 01/09/2026	/reports/source-stock
Buy needed ▲	38.8 t	3 materials are short of open orders	/reports/stock?view=shortfall
Open sales orders	14	212.300 t left to deliver	/sales-orders? status=PENDING,PARTIALLY_SUPPLIED
Delivered this month	₹1.96 Cr	▲ 8% vs Aug (₹1.82 Cr)	/sales?from=2026-09-01&to=2026-09-30

Pick the chart by the question

QUESTION	CHART	DESIGN RULE
Do I have enough of each material for open orders?	Horizontal grouped bars	Sorted by shortage; end labels "Short 24.500 t"; axis starts at 0
How far is this order delivered?	Progress / bullet bar	"18.250 of 30.000 t · 61%"
Are we buying faster than we deliver?	Two-line chart	Direct labels at line ends; solid vs dashed
Are buy and sell rates moving? (later)	Rate trend line	Needs a per-material rates endpoint first
Exact values for many materials	Table with inline bars	Switch automatically when there are more than 8 materials

DO

- Label data directly on bars and line ends
- Use colour only to flag attention (amber/red) plus an icon and words
- Give every number a comparison: "▲ 8% vs Aug"
- Make every card, row and bar a link to its filtered list
- On phones print values on the bars; tooltips don't work on touch

DON'T

- Pie, donut, gauge or 3-D charts
- A legend that relies on colour alone (fails WCAG 1.4.1)
- Animated count-up numbers (slow to read, noisy for screen readers)
- More than 4 KPI cards or more than 2 charts above the fold
- An averaged margin across copper and HMS: it is meaningless

Layout order (see page 7): period control → 4 KPI cards → Needs attention + stock bars → weekly trend + recent activity.
Phones: KPI cards 2 × 2, then Needs attention (most useful in the yard), then charts.

Purchases list: the pattern every list follows

metalix.app/purchases?from=2026-09-01&to=2026-09-30&supplier=shree-ganesh

Purchases

Scrap bought from suppliers. Stock goes up when you save a purchase.

Export

+ New purchase Alt P

Supplier, vehicle or invoice no.

This month · 01/09/2026 – 30/09/2026

Supplier: Shree Ganesh Metals

Material: All

Columns

Comfortable Compact

Supplier: Shree Ganesh Metals X Clear all

6 purchases · 54.195 t · ₹22,15,130.00

Updated 1 min ago

Date	Supplier	Material	Tons (t)	Rate (₹/t)	Amount (₹)	Vehicle	Invoice	
27/09/2026	Shree Ganesh Metals	HMS 1	12.450	34,500.00	4,29,525.00	GJ 03 AX 4521	INV-2291	...
26/09/2026	Shree Ganesh Metals	MS turnings	8.900	29,800.00	2,65,220.00	GJ 03 AX 4521	INV-2287	...
24/09/2026	Shree Ganesh Metals	HMS 1	9.200	34,200.00	3,14,640.00	GJ 03 AX 4521	INV-2280	...
20/09/2026	Shree Ganesh Metals	CRC bundle	14.625	41,000.00	5,99,625.00	GJ 03 BW 7810	INV-2266	...
12/09/2026	Shree Ganesh Metals	HMS 1	7.500	34,000.00	2,55,000.00	GJ 03 AX 4521	INV-2241	...
05/09/2026	Shree Ganesh Metals	Aluminium extrusion	1.520	2,31,000.00	3,51,120.00	GJ 03 BW 7810	—	...

Showing 1–6 of 6

Rows per page 25

< Previous

1

Next >

- 1

One primary action with its shortcut; Export exports the *current filters*.
- 2

Filter bar: search (debounced 300 ms), Indian date presets, supplier, material. Applies instantly on desktop.
- 3

Active filters as removable chips + "Clear all". All filters live in the URL above.
- 4

Summary of the **whole filtered set**, computed on the server (a page-only total would mislead).
- 5

Row just saved: green tint that fades out after 1.5 s; focus moves to it.
- 6

Numbers right-aligned with tabular figures and fixed decimals; units in the header, not the cells.
- 7

Whole row opens the record; other actions in the ... menu (View · Edit · Repeat purchase · Copy link).
- 8

Numbered pages, 25/50/100 rows. No infinite scroll: it loses place on Back and hinders comparing.

Table, filter and search rules

One `DataTable` component (TanStack Table v9, server mode) serves every list. The identifier comes first, numbers line up, and the filter state always lives in the URL.

LIST	COLUMNS, IN ORDER
Purchases	Date · Supplier · Material · Tons (t) · Rate (₹/t) · Amount (₹) · Vehicle · Invoice · ...
Sales Orders	PO no. · Date · Customer · Material · Ordered (t) · Delivered (t) · Left (t) · Progress · Rate (₹/t) · Status · Deliver · ...
Deliveries	Date · PO no. · Customer · Stock from · Material · Tons (t) · Amount (₹) · Vehicle · ...
Companies	Name · Type · GSTIN · Phone · City · ...
Materials	Name · In yard (t) · Open demand (t) · Buy needed (t) · ...

Table rules

- Rows 40 px on desktop (32 px compact option later), 48 px+ on touch.
- Sticky header; row hover tint; long names truncate with a tooltip, **numbers never truncate**.
- Max 1–2 visible row actions; the rest go in the ... menu.
- Status = badge with colour **and** icon **and** word.
- Sorting shown with an arrow and `aria-sort`.
- No bulk selection in v1: nothing here is a batch job except export.
- 6–9 columns visible; the rest behind "Columns".

Filters

- **Desktop:** one toolbar row, filters apply instantly.
- **Phone:** search stays visible; "Filters (2)" opens a bottom drawer with *Apply* and *Reset* (batch filtering suits slow data).
- Sales Orders opens on the **Open** tab, so completed orders never get in the way.
- Zero results keep the header and filters visible and offer "Clear filters".

Date range with Indian FY presets

Today
Yesterday
This week
This month ✓
Last month
This quarter Jul-Sep
This FY 2026-27
Last FY 2025-26
Custom...

< September 2026 >

Mo	Tu	We	Th	Fr	Sa	Su
	1	2	3	4	5	6
7	8	9	10	11	12	13
14	15	16	17	18	19	20
21	22	23	24	25	26	27
28	29	30				

01/09/2026 – 30/09/2026

Each preset shows its resolved dates. FY rule: if today is January–March, the FY started on 1 April of the previous calendar year.

PRESET	ON 27/09/2026
This month	01/09/2026 – 30/09/2026
This quarter	01/07/2026 – 30/09/2026 (FY Q2)
This FY / Last FY	01/04/2026 – 31/03/2027 / 01/04/2025 – 31/03/2026

URL = the view

`/sales-orders?status=PENDING,PARTIALLY_SUPPLIED&material=hms-1` is a real, shareable screen.

Dashboard drill-downs are plain links, Back restores the previous view, and "this month's purchases from Sharma Traders" can be sent on WhatsApp.

Forms: one pattern for every form

Every create/edit form uses the same pattern: shadcn `Field` + React Hook Form, grouped sections, units inside the inputs, a live total, and a Save button that is never disabled.

New purchase ×
Stock goes up when you save

1 ⓘ There is a problem
Enter tons as a number, like 12.450

WHAT & WHO

Date
27/09/2026 📅

Supplier
Shree Ganesh Metals Last used **2**

Material
HMS 1 ⌵

QUANTITY & PRICE

Tons
12,45 t **3**

Rate
₹ 34,500.00 /t **4**

ⓘ Enter tons as a number, like 12.450 Last rate on 24/09/2026

Amount
—
Tons × rate, calculated for you

TRANSPORT & PAPERWORK

Vehicle no.
GJ 03 AX 4521

Invoice no. (optional) **5**
e.g. INV-2291

Cancel Save & add another Save purchase **6**

- 1 Error summary** at the top, focused after a failed submit, linking to each field. Same wording as the inline error.
- 2 Smart defaults:** today, last-used supplier/material, supplier's last rate. Marked "Last used" and always editable.
- 3 Inline error** under the field: red border + icon + words; what the user typed is kept.
- 4 Units inside inputs** (₹, t, /t) via `InputGroup`; numbers right-aligned; widths match the expected value.
- 5** Only **optional** fields are marked; everything else is required.
- 6 Save never disabled.** While saving: spinner + "Saving...", same width, double-submit blocked.

When to validate



React Hook Form: `mode: "onSubmit"`, `reValidateMode: "onChange"`. Limit checks are live warnings (cheap numbers).
Research: on-blur validation cut errors 22% (Wroblewski); GOV.UK says never validate while typing: the hybrid respects both.

Good error copy

"Enter tons as a number, like 12.450"
"Choose a supplier"
"Enter the vehicle number like GJ03AX4521"

Avoid

"Invalid input", "Error 422", "Oops! Something went wrong", "Please enter a valid value": no blame, no codes, no "please/sorry".

New delivery with the live limits panel

Record delivery

PO-0142 · Bharat Castings Pvt Ltd

SALES ORDER

PO-0142

Bharat Castings Pvt Ltd

Change order

Material

HMS 1 (Heavy Melting Scrap)

Rate

₹38,500.00/t

Ordered 30.000 t · Delivered 18.250 t

61%

DELIVERY DETAILS

Date

27/09/2026

Stock from

Shree Ganesh Metals

9,200 t available

Tons

6.000 t

Use max (9.200 t)

Up to 9.200 t

Amount

₹2,31,000.00

6.000 t × ₹38,500.00

TRANSPORT

Vehicle no.

GJ 03 BW 7810

Challan no. (optional)

DC-0352

Cancel

Save & add another

Save delivery Ctrl ↵

Delivery limits

Left on PO-0142

11.750 t

51%

HMS 1 in yard

71.500 t

8%

Shree Ganesh Metals stock

9.200 t

65%

Tightest

You can deliver up to

9.200 t

Limited by Shree Ganesh Metals stock

After this delivery

PO left

5.750 t

Supplier

3.200 t

Yard (HMS 1)

65.500 t

- 1 **Opened from "Deliver"** on PO-0142: the order is locked (customer, material, rate come from it). "Change order" swaps in a picker.
- 2 **"Stock from"** lists only suppliers holding HMS 1, sorted by stock, each showing its tons. Suppliers with 0 t are hidden with a note.
- 3 Quantity **defaults to the maximum allowed**; "Use max" refills it in one tap.
- 4 **Three limits** as meters: typed tons ÷ each limit. The smallest limit is ringed and tagged "Tightest".
- 5 **One sentence** names the binding limit, in plain words.
- 6 **Preview** of what remains after saving: updates as the user types.
- 7 Live amount (tons × rate) announced politely to screen readers.

Backend dependency: a read-only `GET /sales/capacity` endpoint returning the same numbers the server enforces. The limits refetch 300 ms after PO, supplier or date changes.

Limits panel: every state designed

The panel changes colour *and* icon *and* words as the typed tons approach the tightest limit. The server's 409 stays the final guard; when it fires, the panel refreshes from the error's own numbers.

Shree Ganesh Metals stock
9.200 t 65%

You can deliver up to **9.200 t**
Limited by Shree Ganesh Metals stock

1 • Default (< 80%)

Typed 6.000 t. Neutral blue fills.

Shree Ganesh Metals stock
9.200 t 92%

⚠ Close to the limit: 0.700 t to spare

2 • Near limit (80–99%)

Typed 8.500 t. Amber + warning icon; the Tons helper turns amber too.

Shree Ganesh Metals stock
9.200 t ⓧ

Only 9.200 t of HMS 1 from Shree Ganesh Metals is left.

Use 9.200 t Choose another supplier

3 • Over limit (> 100%)

Typed 10.000 t. Red + icon + exact fix. Tons field invalid; on submit the summary appears too.

Checking limits...

4 • Checking

Skeleton rows, only after 500 ms, so fast answers never flicker.

🚫 Limits unavailable

Couldn't check limits right now. We'll still check when you save.

Try again

5 • Unavailable

Network hiccup: the form still works; the server checks on save.

ⓧ Stock changed while you were typing

Only 8.200 t of HMS 1 from Shree Ganesh Metals is left now. Enter 8.200 t or less.

Use 8.200 t

6 • Server 409 after save

Someone else delivered 1.000 t meanwhile. Panel refreshes from `error.details`; no generic toast.

On phones: a strip above the Save bar

Max **9.200 t** · limited by Shree Ganesh stock Details ▲

Save delivery

The panel collapses into a 48 px strip pinned above the sticky Save button. It turns amber or red with an icon and text in the near/over states; "Details" expands the full panel as a second drawer snap point. "Use max" sits inside the Tons input.

Sales order detail: summary first, related records below

metalix.app/sales-orders/0142

Sales Orders > PO-0142

PO-0142

Partly delivered

Bharat Castings Pvt Ltd · HMS 1 (Heavy Melting Scrap) · Ordered on 09/09/2026

<

>

Edit

Record delivery

Alt D

...

Ordered

30.000 t

Delivered

18.250 t

61% of order

Left to deliver

11.750 t

Rate

₹38,500.00/t

Value delivered

₹7,02,625.00

of ₹11,55,000.00

18.250 of 30.000 t delivered

11.750 t left

Deliveries (3)

+ Record delivery

Date	Challan no.	Stock from	Tons (t)	Amount (₹)	Vehicle	
24/09/2026	DC-0340	Shree Ganesh Metals	4.500	1,73,250.00	GJ 03 AX 4521	...
18/09/2026	DC-0327	Jai Ambe Steel	6.250	2,40,625.00	MH 04 KL 2210	...
12/09/2026	DC-0311	Shree Ganesh Metals	7.500	2,88,750.00	GJ 03 AX 4521	...
Total			18.250	7,02,625.00		

Order details

Customer Bharat Castings Pvt Ltd

PO date 09/09/2026

Notes Deliver to Rajkot yard, gate 2

Created Rajesh Patel · 09/09/2026, 11:42 am

Stock for this order

HMS 1 in yard 71.500 t

Jai Ambe Steel 38.000 t

Patel Scrap Co. 24.300 t

Shree Ganesh Metals 9.200 t

See supplier stock →

- 1

Breadcrumb, title + status badge, subtitle with who/what/when.
- 2

Prev/next order, Edit, **one primary action**, ... (Copy link · Print · Cancel order in red).
- 3

3–5 stat cards: the numbers people came for, before the full record.
- 4

Progress strip: "18.250 of 30.000 t delivered · 11.750 t left".
- 5

Related records in the same DataTable used everywhere, with a total row.
- 6

Side cards: order facts and the stock that can serve this order.

Completed: badge "Completed", primary action removed, info alert "All 30.000 t delivered on 26/09/2026." **Cancelled:** a permanent grey alert "This order was cancelled on 14/09/2026 by Rajesh Patel. Deliveries can't be recorded against it."; Edit and Record delivery hidden.

Numbers, dates and complex data

Every number in the app goes through one file, `lib/format.ts`. Exact values in tables and forms; compact values only on summary tiles; never floating-point arithmetic.

KIND	RULE	EXAMPLE
Tons in tables	3 fixed decimals, Indian grouping, unit in the header	1,284.500
Tons in cards, forms, sentences	Same, with " t"	12.450 t
Tons on a KPI tile	1 decimal; exact value in tooltip	286.5 t
Money in tables	2 decimals, lakh/crore grouping, ₹ in the header	11,87,894.21
Money in cards, forms, toasts	With the ₹ symbol	₹11,87,894.21
Rate	₹ with "/t"	₹38,500.00/t
Money on a KPI tile / chart axis	Custom compact: < ₹1 L full rupees; lakh; crore; ≤ 3 significant digits	₹8.75 L · ₹1.96 Cr
Percent	0 decimals in badges and bars	61%
Negative	True minus sign + red + icon	−2.300 t
Zero vs not applicable	Real zero shows zero; not applicable shows a dash	0.000 · —
Record dates	Always DD/MM/YYYY	27/09/2026
Freshness / activity	Relative, exact date on hover	Updated 2 min ago
Change	Arrow + sign + words (never colour alone)	▲ 8% vs Aug

Why a custom compact formatter?

The browser's built-in compact Indian format printed **"1.2KCr"** for about ₹119 crore in testing. Money must never pass through JavaScript floats either: API strings go straight to `Intl.NumberFormat('en-IN')` (which formats strings exactly) and arithmetic like tons × rate uses `big.js`. Dates go to the API as `yyyy-MM-dd`, never through `toISOString()` (which shifts IST dates back a day).

Six ways to make complex data easy

Lead with the answer

Detail pages open with 3–5 stat cards; the full record comes after. Lists show 6–9 columns; the rest are one click away.

Say what it means

"You can deliver up to 9.200 t: limited by Shree Ganesh Metals stock" beats three numbers the user must compare.

Surface exceptions loudly

Shortages, negative stock and aged orders always show colour + icon + words; everything normal stays quiet and grey.

Compare, don't just report

Every KPI carries a comparison ("▲ 31.520 t since 01/09"); progress bars show "18.250 of 30.000 t".

Precision where it's copied

Compact numbers never appear in tables, forms or anything a user might copy into an invoice.

Trade words, one vocabulary

Same labels in lists, forms, badges, toasts and errors. Hindi / Gujarati later through one message dictionary.

Loading and empty states: never a blank or confusing screen

Under 1 s show nothing; skeletons for whole pages (only after 500 ms), spinners for single parts, and old rows stay visible while new ones load. Empty states always say why and offer one way forward.

This month

Date	Supplier	Material	Tons (t)	Amount (₹)

L1 · First load

Header, filters and column titles are live at once; skeleton rows match real row height and column widths. Static or a slow 2 s pulse, no shimmer.

Supplier: Sharma Traders ✕

Updating...

Date	Supplier	Tons (t)	Amount (₹)
27/09/2026	Shree Ganesh Metals	12.450	4,29,525.00
27/09/2026	Sharma Traders	1.275	10,00,875.00
26/09/2026	Patel Scrap Co.	2.140	10,52,880.00

L2 · Filter or page change

Old rows stay at 60% opacity with a 2 px progress line; the summary then reads "12 purchases · 118.400 t". The screen never blanks, even on slow yard data.

⌂ Saving...
✓ Saved

Same width in every state · aria-busy · focus stays on the button · double submit blocked

Dashboard card loading on its own

L3 / L4 · Parts and buttons

A section loads inside its own card only. Save shows a spinner immediately, then a check for 1.2 s, then returns to its label.

No deliveries yet

When scrap goes out to a buyer, record it here. Stock updates automatically.

+ Record delivery
How deliveries work

E1 · First use

Explains what the page is for and gives the one next step. VIEWERs see "Deliveries will appear here once the owner adds them" with no button.

No purchases match these filters

Showing: Copper wire (Berry) · 01/08/2026 – 15/08/2026 · Supplier: Sharma Traders

Clear filters
Change dates

E2 · No results

Shown inside the table body: header and filters stay so the user sees *what* is filtering. Search variant: "No results for 'bras'. Check the spelling, or search by GSTIN."

BUY NEEDED

✓ **All materials are stocked.**

Nothing needs to be purchased for open orders right now.

E4 · All done

A positive empty state with no action. Never guilt-inducing ("You haven't..."). Empty states render only after a successful load with zero rows, never during loading or after an error.

Errors, success and confirmations

Errors sit next to their cause, in plain words, and keep what the user typed. Success is shown in the page first; the toast only confirms. Confirmations are rare and name the action.

 **Purchase not saved**
 Something went wrong on our side and this purchase was not saved.
 Try again. If it keeps happening, share this reference with support.
 Reference: 7f3a9c2e 
 Try again Copy reference

Er4 · Server error (500)

Persistent inline alert in the form, never a vanishing toast. The requestId is the only technical token users ever see.

 **You're offline. Changes can't be saved until the connection is back.**
 Page content stays usable for reading; the form keeps its entries.

Your session has ended
 Log in again to save this delivery. Your entries are kept.
 Password
 Log in

Er5 / Er7 · Offline and session ended

A non-dismissible banner while offline ("Back online" for 3 s after). An expired session opens a small login dialog over the form instead of redirecting.


We couldn't find this sales order
 The link may be wrong, or the order number was typed incorrectly.
 Go to Sales Orders

Stock by material
 Couldn't load stock by material. Check your connection and try again.
 Try again

Er6 / Er1 · Not found and section error

A failed section fails alone, inside its card, with its own "Try again"; the rest of the dashboard keeps working.

 **Cancel order PO-0151?**
 Kutch Foundry Works ordered 20.000 t of Brass honey.
 Nothing has been delivered yet. After cancelling, no deliveries can be recorded against this order.
 Keep order Cancel order

C1 · Confirm a serious action

Names the record and the consequence; buttons are verbs ("Keep order" / "Cancel order", never "Cancel" / "OK"); focus starts on the safe button.

 **Delivery saved** 
 6.000 t of HMS 1 to Bharat Castings Pvt Ltd
 View Add another

27/09/2026	DC-0352	Shree Ganesh Metals	6.000
24/09/2026	DC-0340	Shree Ganesh Metals	4.500

S1 · Success

The sheet closes, the new row is tinted and focused, *then* a toast confirms. Toasts with actions stay 10 s with a close button; plain ones 4 s.

 Copied icon swaps for 1.5 s, "Link copied" announced
 Saved inline save confirmation, 1.2 s
 All 30.000 t delivered on 26/09/2026.

S2 / S3 · Small confirmations

Tiny successes don't need a toast: the control itself confirms. Persistent facts use an in-page Alert with icon + word, not only colour.

Every API error, mapped to the screen

One dictionary keyed by `error.code` turns every failure into plain words that reuse the server's own numbers, plus a one-tap fix.

BACKEND RESPONSE	WHERE IT APPEARS	WHAT THE USER READS	ONE-TAP FIX
422 field issues	Under each field + focused "There is a problem" summary	"Enter tons as a number, like 12.450" · "Choose a supplier"	Summary links jump to fields
409 PO_QUANTITY_EXCEEDED	Tons field + alert; limits panel refreshed	"PO-0142 has only 11.750 t left to deliver. You entered 12.000 t."	Use 11.750 t
409 INSUFFICIENT_STOCK	Same	"Only 71.500 t of HMS 1 is in the yard. Enter 71.500 t or less."	Use 71.500 t
409 INSUFFICIENT_SOURCE_STOCK	Same + supplier field highlighted	"Only 9.200 t of HMS 1 from Shree Ganesh Metals is left. Enter 9.200 t or less, or choose another supplier."	Use 9.200 t · Choose another supplier
409 DUPLICATE_VALUE	The offending field	"Invoice number ST/26-27/418 is already used on another purchase."	Open that purchase
409 NEGATIVE_STOCK_HISTORY	Alert inside the edit form	"This change would make HMS 1 stock go below zero on 18/09/2026. Change the quantity or date, or edit the later deliveries first."	View stock history
403	Prevented by hiding actions; inline alert if it happens	"You have view-only access. Ask the owner if you need to make changes."	—
404	Full-page state	"We couldn't find this sales order. The link may be wrong."	Go to Sales Orders
401	Silent refresh; else login dialog over the form	"Your session has ended. Log in again to save this delivery. Your entries are kept."	Log in
429	Non-blocking warning	"Too many requests in a short time. Wait a few seconds and try again."	Try again
500	Persistent inline alert	"Something went wrong on our side and this purchase was not saved..." + Reference	Try again · Copy reference
Network / offline	Inline + global banner	"No internet connection. Your entries are still here. Try again when you're back online."	Try again
Render crash	Route / widget error boundary	"This page stopped working. Reload the page. Your saved records are safe."	Reload

Rules for all messages

- Say what happened and how to fix it; never show raw codes.
- No "invalid", "please", "sorry", "oops"; no blame.
- Errors use `role="alert"`; results use `role="status"` (WCAG 4.1.3).

Confirm only these four

- Cancelling a sales order
- Deactivating a company or material
- Saving a backdated edit that changes stock history (impact shown first)
- Leaving a form with unsaved changes

Micro-interactions and motion

A micro-interaction is a trigger plus its feedback. Motion exists only to give feedback, show a change of state, or show where something came from: subtle, brief, and never in the way of typing.

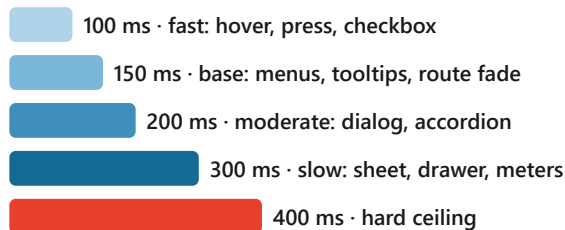
KEEP THESE SIX

- **Press:** buttons shift 1 px down; no ripple
- **Save:** spinner → check for 1.2 s → label
- **Meters:** progress and limit fills glide to the new value (300 ms)
- **Copy:** icon swaps to a check + "Copied" for 1.5 s
- **Just saved:** new/edited row tints green, fades over 1.5 s
- **Overlays:** shadcn's default menu, sheet, dialog, accordion, tab motion

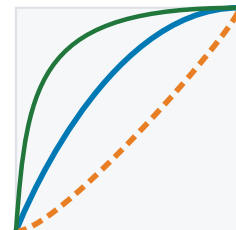
REJECT THESE

- KPI count-up numbers (delay reading the real value)
- Shaking fields on error (vestibular trigger; never the only signal)
- Shimmer skeletons (use static or slow pulse)
- Confetti, parallax, pull-to-refresh, animated illustrations
- Animation on typing, keyboard focus, or re-sorting large tables

Duration and easing tokens



0 400 ms (500 ms = "a real drag")



Enter
(0, 0, .38, .9)
Exit
(.2, 0, 1, .9)
Sheets (M3)
(.05, .7, .1, 1)

Enter decelerates; exits are faster (0.7–0.8× the entrance).

ELEMENT	WHAT MOVES	ENTER	EXIT	REDUCED MOTION
Button hover / press	colour; press translateY(1px)	100 ms	100 ms	colour only
Tooltip, menu, popover, combobox	opacity + scale 0.95→1	150 ms	100 ms	fade 100 ms
Dialog / AlertDialog	backdrop fade; panel fade + scale	200 ms	150 ms	fade; no blur on phones
Sheet (right) / Drawer (bottom)	slide in from the edge	300 ms	200 ms	fade 100 ms
Progress / limit meter	scaleX to new value	300 ms	—	instant
Route change	React <code><ViewTransition></code> cross-fade	150 ms	150 ms	none
New-row highlight	background colour (not motion)	instant	1.5 s	same

Performance rules for low-end Android

Animate only `transform` and `opacity`; remove shadcn's backdrop blur on phones; honour `prefers-reduced-motion` **and** offer a "Reduce animations" switch in the user menu. Feedback must paint in the very next frame (INP ≤ 200 ms) or users click twice.

Hover, focus, active, disabled and other states

One state vocabulary for every control. Focus is always a solid 2 px ring with a 2 px gap (not shadcn's 50% transparent ring, which can miss the 3:1 contrast rule).

	DEFAULT	HOVER	FOCUS (keyboard)	PRESSED	LOADING	DISABLED*
Primary button						
Outline button						
Tons input						
Table row						
Sidebar item						

* "Pressed" for the sidebar item shows the **selected** state (tint + bold + 3 px bar + `aria-current="page"`). Disabled is avoided wherever possible; when unavoidable use `aria-disabled` + a visible reason, because a disabled button can't show a tooltip or receive focus.

Invalid

Tons

ⓘ Enter 9.200 t or less

Red border + soft red ring + icon + message linked with `aria-describedby`; value kept.

Read-only (VIEWER, locked PO)

Sales order

Muted background + lock icon; still focusable and copyable; no hover or focus border change.

RULE	VALUE
Focus ring	<code>outline-2 outline-offset-2 outline-ring, :focus-visible</code> only; $\geq 3:1$ contrast
Hover	Pointer devices only; 100 ms colour change; never the only way to find information
Selected	Colour plus a non-colour cue (check, bar, bold) and the right ARIA attribute
Cursor	Restore <code>cursor-pointer</code> on buttons (Tailwind v4 changed the default)

Responsive rules and breakpoints

One codebase, three shapes. The main switch is 768 px (shadcn's mobile breakpoint): below it, navigation moves to the bottom, tables become cards and dialogs become bottom drawers.

	640	768 ♦ main switch	1024	1280
Phones	Large	Tablets	Laptops	Desktops
bottom nav · cards drawers · sticky Save		icon-rail sidebar · table scrolls, first column pinned	full sidebar · sheets detail 2/3 + 1/3	4-up KPI grid full-width lists
ELEMENT	DESKTOP	PHONE (< 768 PX)		
Navigation	Sidebar (256 px, collapsible)	Bottom bar: Home · Purchases · + · Orders · More		
Lists	DataTable, numbered pages	Tappable cards, "Load more"		
Create / edit	Right Sheet	Full-height bottom Drawer, Save bar pinned above the keyboard		
Dialogs, confirmations	Centred Dialog	Bottom Drawer, 48 px buttons, safe option on the left		
Filters	Toolbar, instant	"Filters (2)" drawer with Apply / Reset		
Dashboard	4 KPI cards in a row, charts beside lists	KPIs 2 × 2, then Needs attention, then charts		
Charts	Hover tooltips	Values printed on bars (no tap tooltips)		

Touch and typing on phones

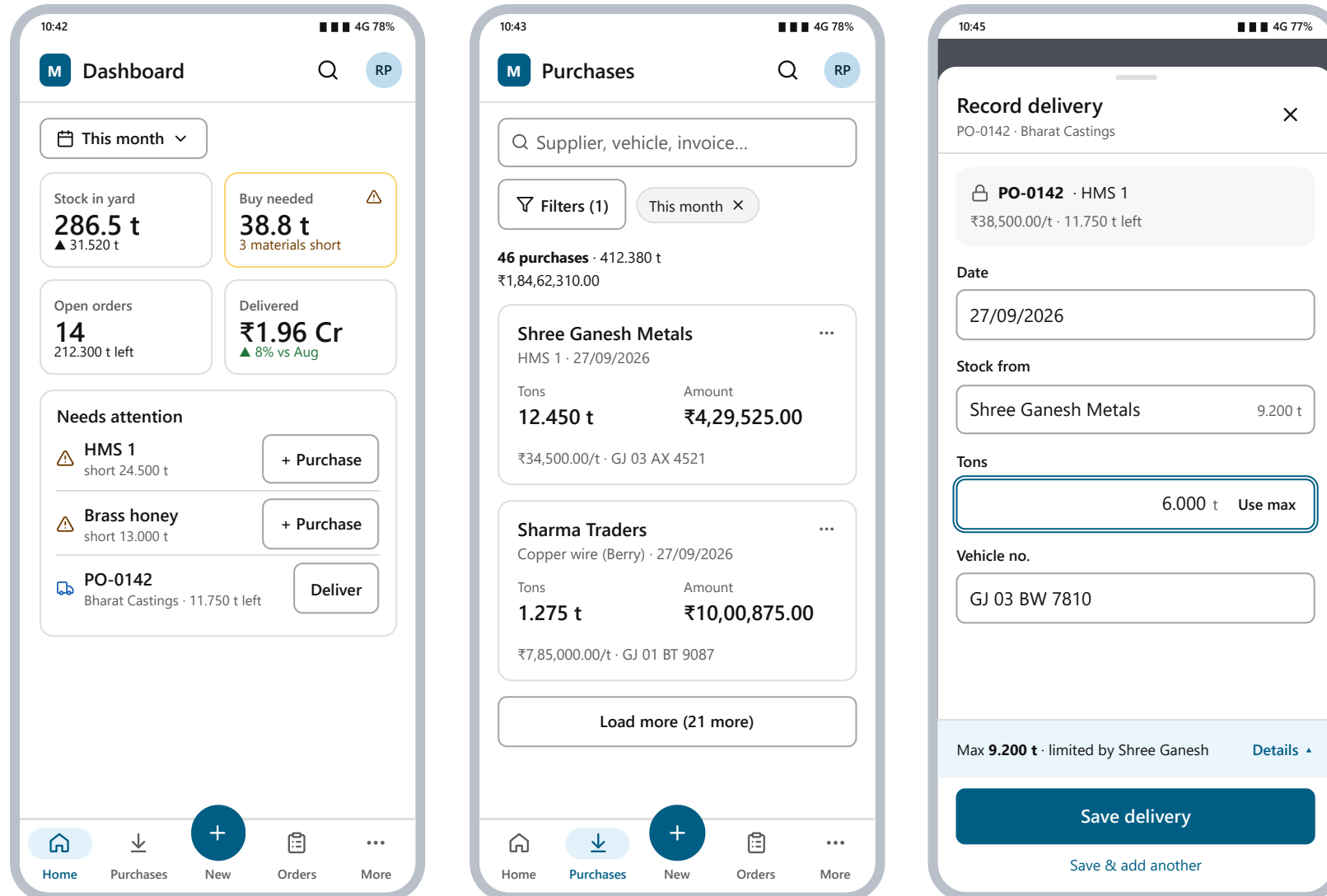
- **Targets:** WCAG minimum 24 px; the app uses **44 px** buttons and inputs, 48 px list rows and 64 px bottom-bar items on touch (`pointer-coarse: variant`); desktop keeps shadcn's density.
- **16 px** input text so iPhones don't zoom.
- Numbers: `inputmode="decimal"` ; phones: `type="tel"` ; vehicle and GSTIN: capitals automatically.
- Keyboard "Next" key moves between fields; "Done" on the last one.
- Primary action in a sticky bottom bar, clear of the phone's gesture area (`safe-area-inset`).
- Destructive actions live in ... menus, never next to Save.
- Things used most in the yard (the "+" button, Save) sit in the thumb zone at the bottom.
- No pull-to-refresh, no swipe-only actions: every gesture has a button alternative.

"Sunlight" theme for outdoor use

Standard Stock in yard 286.5 t ▲ 31.520 t since 01/09 Tons	Sunlight Stock in yard 286.5 t ▲ 31.520 t since 01/09 Tons
--	--

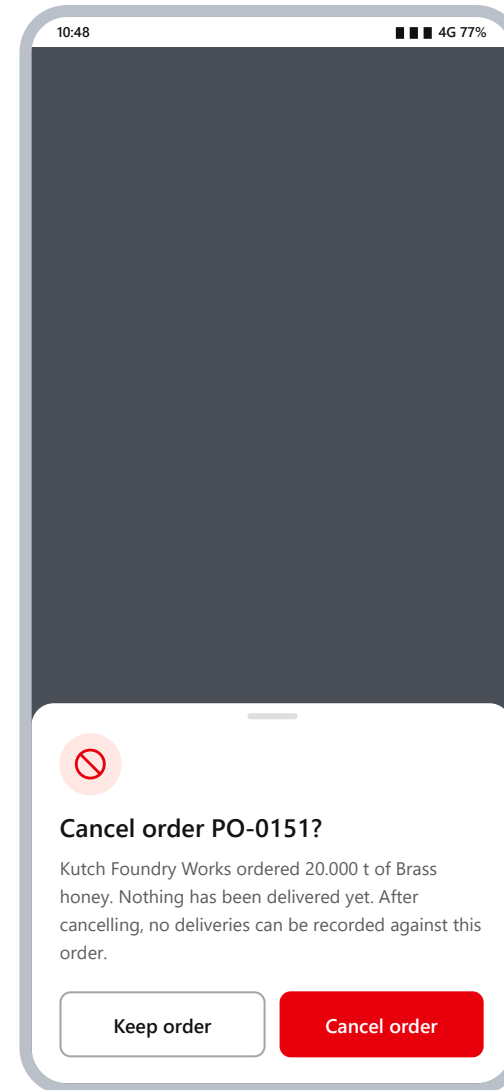
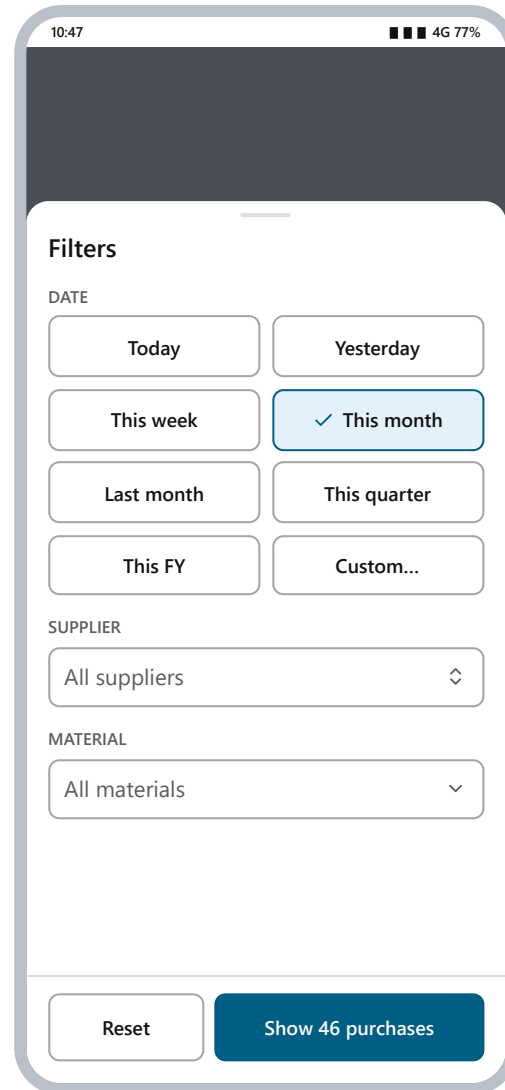
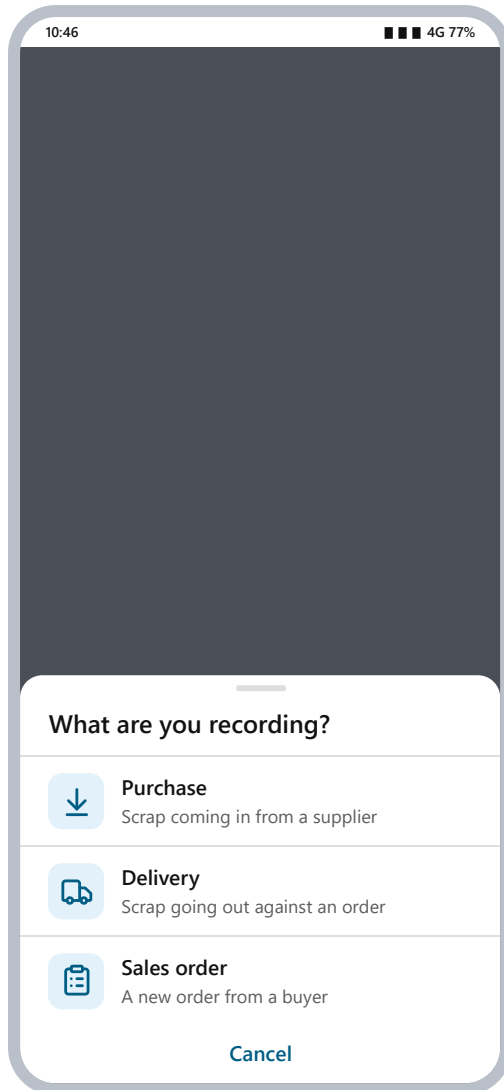
Pure white background, near-black text at 7:1 or better, 2 px input borders, no faint greys, bold key figures, 16 px base font. Switched from the user menu; also follows the phone's "increase contrast" setting.

Phone screens (360 × 800, a typical mid-range Android)



- Dashboard:** KPIs as a 2 × 2 grid; "Needs attention" right under them with 44 px action buttons.
- Bottom bar:** 4 destinations + a raised centre "+" in thumb reach. Active item: tinted pill, blue icon and bold label.
- Lists:** sticky 44 px search, a "Filters (1)" button, the summary, then cards: identifier, 2 key numbers, secondary line.
- Forms:** full-height drawer, 44 px fields with 16 px text, "Use max" inside the Tons field.
- Limits strip** pinned above the Save button; "Details" expands the full panel.

Phone drawers: create, filter, confirm



- 1 **"+" drawer** asks one plain question. Three 64 px rows, each with an icon, a name and what it means.
- 2 **Filter drawer** (batch filtering): preset chips in a 2-column grid, then pickers. The main button shows the live result count.
- 3 **Confirmations** become bottom drawers with two 48 px buttons: the safe option on the left, the destructive one on the right.

Drawers slide up in 300 ms with a decelerating curve and leave in 200 ms. A visible handle and a close button: swipe-down is never the only way to dismiss.

Accessibility checklist (WCAG 2.2 AA)

Base UI gives keyboard support, focus trapping and ARIA roles for free. The items below are the app's own responsibility; most are token and wiring work, done once in shared components.

CRITERION	WHAT THE APP MUST DO	WHERE
1.4.3 Text contrast	≥ 4.5:1 for all text; darken muted text so it passes on grey panels; dark text on amber badges	Tokens
1.4.11 Non-text contrast	Input borders, checkboxes, focus ring and chart marks ≥ 3:1	Tokens
1.4.1 Use of colour	Badges, deltas, meters and chart series always carry text or an icon too	Components
2.1.1 Keyboard	Every action reachable with Tab / Enter / Space / Esc; rows focusable; skip link	Components
2.4.7 / 2.4.11 Focus visible, not hidden	Solid 2 px ring; scroll padding so sticky header, Save bar and bottom nav never cover the focused field	Global CSS
2.4.3 Focus order	Focus returns to the trigger after overlays close; moves to the new row after save	Components
2.5.8 Target size	≥ 24 px everywhere; 44–48 px on touch	Tokens
2.5.7 Dragging	Any drag (reorder, upload) has a button alternative	Features
3.3.1 / 3.3.3 Errors	Text errors with a fix, <code>aria-invalid</code> , <code>aria-describedby</code> , focused summary, hidden "Error:" prefix	Forms
3.3.4 Error prevention	Limits panel before submit; confirm or undo for cancel, deactivate, backdated edits	Features
4.1.3 Status messages	Toasts, "Saved", "Copied", "Showing 24 purchases" in polite live regions; errors as alerts	Components
3.2.6 Consistent help	Help, shortcuts and WhatsApp support in the same place on every page	Shell
3.3.7 Redundant entry	Pre-fill supplier, material, rate, PO data and vehicle from earlier steps	Forms
3.3.8 Accessible login	Allow paste and password managers; <code>autocomplete</code> on login fields	Login
1.3.1 Structure	Real tables with header cells and <code>aria-sort</code> ; cards use description lists; every input has a label	Components
1.1.1 Non-text content	Charts have a text summary and "View as table"; icon-only buttons have labels	Components
1.4.4 / 1.4.10 Zoom and reflow	Usable at 320 px width and 200% zoom; never block pinch-zoom	Global
3.1.1 Language	<code><html lang="en-IN"></code> ; <code>lang="hi"</code> / <code>"gu"</code> on translated text later	Shell
2.3.3 Motion (AAA, cheap)	Reduced-motion rule covers all animation; in-app "Reduce animations" switch	Global CSS

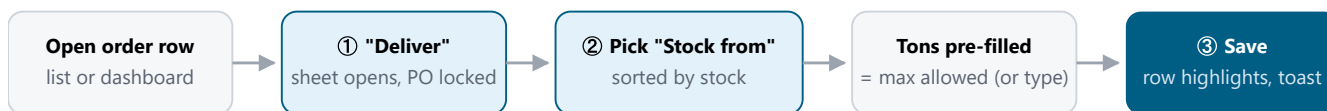
Keep it enforced

Check every colour pair in tweakcn's contrast checker before use, run `eslint-plugin-jsx-a11y` in CI, and test once with a keyboard only and once with Android TalkBack before each release.

User flows and click counts

Target: three clicks per core task, with defaults doing the rest. A "click" includes choosing an option; typing is not counted.

Record a delivery against an order: 3 clicks



Keyboard: Alt+D → pick PO → pick supplier → Ctrl+Enter · Phone: 4 taps

Record a purchase: 4 clicks (2 when defaults match)



FLOW	FASTEST START	DESKTOP CLICKS	PHONE TAPS
Record a purchase	+ New > Purchase, or "+ Purchase" beside a shortage	4 (2)	5
Create a sales order	Sales Orders button, customer page, Alt+O	4	5
Record a delivery	"Deliver" on the order row	3	4
Check what to buy	Dashboard "Buy needed"	0–1	0–1
Look up a company's history	Ctrl+K → name → Enter	2	3
See a supplier's stock	Sidebar > Supplier stock	2	3

The click reducers, by impact

- **"Deliver" where the decision is made:** no PO picking
- **Quantity defaults to the maximum;** "Use max" in every limit error
- **Predictable defaults,** tagged "Last used", always editable
- **Inline "+ Add '...'"** in every picker (Alt+C, as in Tally)
- **"Save & add another"** keeps date and supplier for the next truck
- **"Repeat purchase"** in the row menu for regular suppliers
- **Filters in the URL:** no re-filtering after Back
- **Sales Orders opens on "Open"**
- **Keyboard-first:** Enter to next field, Ctrl+Enter to save
- **Centre "+" on phones:** any transaction in two taps

Which shadcn/ui component for which job

Decide once at `shadcn init` : **Base UI** (the default since July 2026, with native Combobox and Drawer), compact Nova style. Never mix Radix and Base UI. Borrow layout ideas from shadcn-admin; don't fork it (it is Radix-based).

NEED	SHADCN/UI COMPONENTS	WRAPPED AS
App shell	Sidebar (icon-collapsible, inset), SidebarTrigger, SidebarMenuBadge, Breadcrumb, DropdownMenu, Avatar	AppShell, AppSidebar
Mobile navigation	Button + router links, Sheet ("More"), Drawer ("+")	BottomNav
Command palette	Command in CommandDialog, CommandShortcut, Kbd	CommandPalette
Page header	Breadcrumb, Button, ButtonGroup, DropdownMenu	PageHeader
KPI tiles	Card (Header/Title/Action/Content/Footer), Badge, Tooltip, Skeleton	StatCard
Charts	Chart (ChartContainer, ChartTooltip, accessibilityLayer), ToggleGroup	—
Data tables	Table + TanStack Table v9 guide, DropdownMenu, Pagination, Select, ScrollArea	DataTable
Mobile lists	Item / ItemGroup or Card	DataTable renderMobileCard
Filters and search	InputGroup, Popover + Calendar (range), Combobox, Select, Tabs, Badge, Drawer	FilterBar, DateRangePicker
Forms	Field, FieldLabel, FieldDescription, FieldError, FieldSet, FieldLegend, Input, InputGroup, Textarea, RadioGroup, Switch	FormField, FormSection, ErrorSummary
Company / material pickers	Combobox (Base UI) with async search + Dialog	EntityCombobox
Money and tons inputs	InputGroup + react-number-format (lakh grouping)	MoneyInput, TonsInput
Delivery limits	Card, Progress, Alert, Button	LimitMeter, DeliveryLimitsPanel
Status	Badge with added success / warning / info variants	StatusBadge
Create / edit containers	Sheet (desktop), Drawer (phone), Dialog (inline create)	FormSheet, ResponsiveDialog
Confirmation	AlertDialog	ConfirmDialog
Loading	Skeleton, Spinner, Progress	—
Empty states	Empty (EmptyMedia, EmptyTitle, EmptyDescription, EmptyContent)	EmptyState
Messages	Alert (+ variants); Sonner behind one wrapper	lib/notify.ts
Help	Tooltip (icon buttons only), Popover (info "i"), Kbd	—
Detail pages	Card, Tabs, Item, Separator, Badge, Progress	DetailHeader, KeyValueType
Login	login-03 block (Card + Field + Input)	—

Pin these to avoid known breakages

`react-day-picker` (shadcn's Calendar breaks on DatePicker v10 until regenerated), `recharts@3` , `@base-ui/react` . Run `shadcn add --diff` before every component update.

Design tokens: colour, type, spacing

Components use only semantic tokens (`bg-success` , never `bg-green-500`). Base colour is Neutral; only the primary steel blue carries a hue. **Values are proposals:** check each pair's contrast in tweakcn before use.

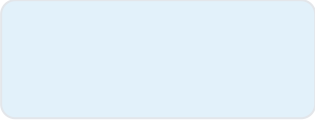
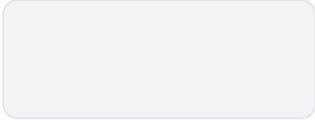
 primary oklch(0.45 0.10 235) buttons, active nav, focus	 accent oklch(0.95 0.02 235) hover, selected rows	 muted / muted-fg 0.967 / 0.48 (darkened) panels, secondary text	 input border oklch(0.62 0 0) ≥ 3:1 on white
 success oklch(0.50 0.12 150) completed, saved	 warning oklch(0.84 0.16 84) pending, near limit, offline	 info oklch(0.52 0.15 255) partly delivered, tips	 destructive oklch(0.577 0.245 27.3) over limit, errors, cancel

Chart series      blue / orange for two-series charts; ≤ 4 series per chart

Type scale (Inter + Noto Sans Devanagari/Gujarati)

286.5 t KPI 30/36

Sales Orders Title 24/32

Needs attention Section 18/28

Stock in yard Label 14/20 · 500

Bharat Castings Pvt Ltd Body / cell 14/20
Updated 2 min ago · Meta 12/16

Phones: body and inputs 16 px. Devanagari/Gujarati need looser leading (≈ 1.6). `tabular-nums` on every number.

Spacing (4 px base) and shape

Page gutter	16 / 24 px	
Between sections	24 px	
Card padding	16 / 24 px	
Between form fields	16 px	
Label → input	8 px	
Table row	40 / 48 px	
Radius	0.5rem	controls 8 px · cards 11 px
Elevation	flat	1 px borders; shadows only on menus/dialogs

Reusable components and folder structure

Three layers: shadcn output (treated as vendored code), the app's own composite components, and feature screens. Feature code never imports Base UI or the toast library directly.

```
src/
  components/
    ui/          shadcn CLI output only
    layout/
      app-shell  app-sidebar
      site-header  bottom-nav
      command-palette  shortcuts
    common/
      page-header  detail-header
      data-table/  (table, toolbar,
        pagination, skeleton,
        summary-strip)
      filter-bar/  (filter-bar,
        date-range-picker, chips)
      form/  (form-sheet, form-section,
        form-field, error-summary,
        entity-combobox, money-input,
        tons-input, date-input,
        vehicle-input)
      stat-card  status-badge
      limit-meter  empty-state
      error-state  confirm-dialog
      responsive-dialog
      key-value-list  freshness
      offline-banner
    features/
      dashboard  purchases
      sales-orders  deliveries
      companies  materials
      reports  auth
      └─ api/ schemas/ components/
    lib/
      api-client  errors  format
      decimal  fy  notify
      permissions  timing
    hooks/  styles/
```

COMPOSITE	KEY PROPS
PageHeader	title, description, primaryAction, secondaryActions (≤ 3), breadcrumbs
DataTable	columns, data, rowCount, state, renderMobileCard, emptyState, isLoading, isPlaceholderData, error, getRowHref, highlightRowId
FilterBar	typed filter config; reads/writes URL search params
DateRangePicker	value, onChange, presets (Indian FY)
EntityCombobox	fetcher, filter (e.g. SALE/BOTH), onCreate (" + Add"), renderOption
MoneyInput / TonsInput	string in, string out ("30.250"); max + "Use max"
StatCard	label, value, exactValue, delta, footnote, href, loading
StatusBadge	PENDING · PARTIALLY_SUPPLIED · COMPLETED · CANCELLED · ACTIVE · INACTIVE
LimitMeter	label, value, limit, isBinding, state
EmptyState	kind: first-use · no-results · all-done
ConfirmDialog	<code>await confirm({ title, body, confirmLabel, cancelLabel, destructive })</code>
FormSheet / ResponsiveDialog	Sheet or Dialog on desktop, Drawer on phones; unsaved-changes guard

Consistency guard-rails

- Edits to `components/ui` are limited to added variants and accessibility fixes, logged in `CHANGES.md` ; updates via `shadcn add --diff` .
- Lint bans raw colour classes (`bg-red-500`) and direct imports of `@base-ui/react` / `sonner` in features.
- A dev-only `/dev/components` page shows every composite in every state: cheaper than Storybook for a small team.
- All numbers through `lib/format.ts` ; all toasts through `lib/notify.ts` ; all permissions through `can('write')` .

What to build first: 20 priorities

Ranked by how much friction each removes. P0 items make the core daily tasks safe and fast; P1 makes them pleasant; P2 adds polish and insight.

RANK	CHANGE	WHY IT MATTERS	EFFORT
P0-1	Live delivery limits panel with "Use max" and the binding-limit sentence	Turns the strictest rejections into guidance (needs capacity endpoint)	M
P0-2	"Deliver" on open order rows, order page and dashboard, order pre-locked	Delivery in 3 clicks; no wrong-PO errors	S
P0-3	Smart defaults: today, last supplier / material / rate, max quantity	Most users keep defaults; purchase in 2 clicks	S
P0-4	One <code>lib/format.ts</code> : exact en-IN values, custom lakh/crore compact	No misread rupee figures, no "1.2KCr"	S
P0-5	Plain-language error dictionary with one-tap fixes; 422 → field + summary	Users fix errors without help	S
P0-6	Keep previous rows while loading; skeleton only on first load after 500 ms	Screen never blanks on weak data	S
P0-7	URL-held filters with Indian FY presets	Shareable views, working Back	M
P0-8	Hide write actions for VIEWER; "View only" badge	No dead ends or 403s	S
P0-9	Wait-for-server saves on stock/money, then highlight row + short toast	No numbers that roll back after the user saw them	S
P1-10	EntityCombobox with inline "+ Add" (Alt+C)	No leaving the form to create masters	M
P1-11	"Save & add another" and "Repeat purchase"	Fast batch entry at the weighbridge	S
P1-12	Ctrl+K palette, Alt shortcuts, Enter-to-next, Ctrl+Enter save	Tally-speed entry for accountants	M
P1-13	Bottom nav with centre "+", drawer forms, sticky Save, 44–48 px targets	One-handed use outdoors	M
P1-14	Sunlight theme, stronger input borders, darker muted text	Outdoor readability, AA contrast	S
P1-15	Form drafts saved locally + re-login dialog that keeps entries	No lost work on dropped connections	S
P1-16	Dashboard "Needs attention" with inline actions and drill-downs	"What to buy / deliver" at a glance	M
P2-17	Column visibility and density remembered per user	Power users see more	S
P2-18	Recent items in the palette and sidebar	Faster return to working records	S
P2-19	Rate trend and margin charts	Pricing insight (needs rates endpoint)	M
P2-20	Component showcase page, lint guard-rails, Hindi/Gujarati	Keeps the system consistent as it grows	M

Suggested order of work

1 · Foundation

init, tokens, shell, api-client

2 · Composites

DataTable, FilterBar, form kit

3 · Delivery + one list

test with 3–5 real users

4 · Remaining screens

dashboard, reports, masters

Open questions and next steps

Most of this guide rests on established research and design-system guidance. A few choices are reasoned rather than tested, and a few need small backend additions.

Backend additions the UI depends on

ENDPOINT	UNLOCKS	PRIORITY
GET /sales/capacity? poId&sourceCompanyId&saleDate&excludeSaleId	The live limits panel with the same numbers the server enforces	Highest
Last rate for a supplier + material	The "last rate" default on purchases	High
GET /reports/rates (day / week / month buckets)	Rate trend chart; KPI comparisons with the previous period	Medium
Refresh token in an httpOnly cookie	Safer sessions (token never readable by page scripts)	Medium
Idempotency key on create endpoints	Safe "Try again" on flaky yard connections without duplicates	Medium

Validate with real users before building everything

- Run a short session with **3–5 people** (owner, accountant, yard staff) on a **mid-range Android phone in daylight**, using the delivery form and one filtered list.
- Check: click counts, whether hiding (rather than disabling) buttons for VIEWERs confuses anyone, and whether the trade words ("Left to deliver", "Stock from") read naturally.
- Confirm the Sunlight theme is readable outdoors and the 44 px targets are comfortable with work gloves.

Still to verify

- Contrast of every proposed colour pair (tweakcn) in light, dark and Sunlight themes.
- Vehicle-number and GSTIN patterns against official formats.
- That the shadcn data-table guide and chosen community table code work with TanStack Table v9.

Main sources behind this guide

Nielsen Norman Group · GOV.UK Design System · Baymard Institute · Shopify Polaris · IBM Carbon · GitHub Primer · Material 3 · Microsoft Fluent 2 · W3C WCAG 2.2 · web.dev · MDN · shadcn/ui, Base UI, TanStack and React 19.3 docs · Stephen Few · UK Government Analysis Function · Tally, Vyapar and Khatabook patterns. Full citations: companion report *shadcn UI UX design system.md*.

In one sentence

Build one calm, number-first interface where the three limits behind every delivery are visible before the user acts, every screen follows the same patterns, and every interaction gives quick, plain feedback, on a desk PC and on a phone in the yard.